

Ali Raza

Full-Stack AI Engineer

Multan, Pakistan | +92 324 5892281 | aliraza.ai.dev3@gmail.com
ali-raza-engineer.vercel.app | github.com/AliRaza-Dev678 | linkedin.com/in/ali-raza-597b013b0

PROFESSIONAL SUMMARY

Full-Stack AI Engineer and Software Engineering undergraduate building intelligent products across React/TypeScript and Next.js frontends, FastAPI services, PostgreSQL/Redis data layers, LLM and agentic workflows, and n8n / Make.com / GoHighLevel automation pipelines. Designed a containerized AI-powered workspace platform with secure authentication, role-aware access, real-time WebSockets, and Groq-powered insights, and architected a reusable GoHighLevel CRM snapshot that deploys a full lead-to-close system for real estate clients in under an hour. Further experience spans document and video RAG (Self-RAG and CRAG patterns), LangGraph agents, LangSmith tracing, Flutter mobile development, NLP, and computer vision, with a focus on turning AI prototypes into complete, usable software systems.

TECHNICAL SKILLS

Full-Stack & Platform Engineering: Python, FastAPI, React, Next.js, TypeScript, JavaScript, Vite, Tailwind CSS, Flutter, Dart, REST APIs, WebSockets, JWT/OAuth2, role-aware authorization, Pydantic

Generative AI & LLMs: RAG pipelines, Self-RAG, CRAG, agentic systems, prompt engineering, embeddings, semantic search, MMR retrieval, LangChain, LangGraph, LangSmith, ReAct, Groq API, Anthropic and OpenAI models, Hugging Face, Llama models

Data, Persistence & Background Processing: PostgreSQL, PGVector, Redis, Celery, Tortoise ORM, Aerich, SQLAlchemy, SQLite

Machine Learning, NLP & Vision: Scikit-learn, TensorFlow, PyTorch, Pandas, NumPy, TF-IDF, sentiment analysis, classification, clustering, SVM, Random Forest, Decision Trees, K-Means, Naive Bayes, regression, CNNs, LSTMs, transfer learning, ResNet50

Workflow Automation & Deployment: n8n, Make.com, GoHighLevel (CRM architecture, funnels, workflows, Conversation AI, A2P 10DLC compliance), Docker Compose, Vercel, Streamlit, Git, GitHub, Jupyter Notebook, Google Colab, Kaggle, OpenAPI/Swagger

SELECTED PROJECTS

Reusable GoHighLevel Real Estate CRM & Automation Snapshot [Docs] *GoHighLevel, Conversation AI, HTML email, A2P 10DLC*

- Architected a client-agnostic GoHighLevel snapshot covering the full lead-to-close lifecycle: 14 workflows, 53 custom fields, 41 tags, 3 pipelines, 4 calendars, 6 landing pages, and 32 SMS/email templates.
- Built deployment around 30 Custom Values so every asset rebrands from a single configuration table, cutting per-client onboarding from a day of manual edits to under an hour.
- Configured a Conversation AI agent for inbound qualification and calendar booking with defined escalation rules, alongside sub-60-second speed-to-lead, lead scoring, no-show recovery, and 45-day buyer/seller nurture sequences.
- Engineered compliance into every workflow: A2P 10DLC-ready consent language, DND checks before every send, and exit conditions on all automated sequences.

Intelligent AI Assistant & Content Automation Pipelines [Code] *n8n, Make.com, Groq, LangChain, Google Workspace APIs, Meta API*

- Engineered a Personal Assistant Agent within n8n, using Groq and LangChain to automate Google Calendar scheduling, Gmail management, Google Docs generation, and Google Sheets expense tracking.
- Developed a modular content distribution engine polling Google Sheets for state management and auto-publishing media to YouTube and Instagram Reels via the Meta Graph API.
- Implemented asynchronous branching, parallel execution tracks, and web search integration for multi-step agents.

AI-Powered Workspace Management Platform [Live | Code] *Python, FastAPI, React, PostgreSQL, Redis, Celery, Docker*

- Designed a full-stack collaboration platform for workspaces, teams, projects, tasks, files, and analytics.
- Developed secure JWT authentication, role-aware authorization, real-time WebSockets, and Groq-powered insights.
- Added Redis caching, Celery background reports, and CSV exports; containerized services with Docker Compose.

AI Chat Assistant (ChatGPT Clone) [Live | Code] *React, TypeScript, FastAPI, LangGraph, PostgreSQL, Docker*

- Built a production-style assistant with React/TypeScript, FastAPI, and LangGraph serving Groq-hosted gpt-oss-120b.
- Implemented token-by-token streaming, persistent conversations, stop/regenerate controls, and Markdown rendering.

IUB University & YouTube RAG Chatbots [Code] *LangChain, PGVector, PostgreSQL, Hugging Face, Groq, Streamlit*

- Built RAG assistants to ingest official IUB PDFs and YouTube transcripts (via yt-dlp), storing vectors in PGVector.
- Orchestrated grounded answer generation with Llama 3.1 via Groq and MMR retrieval behind Streamlit interfaces.

Additional Projects [Portfolio | Repos] *Next.js, Tailwind CSS, LangGraph, FastAPI, Tortoise ORM, Scikit-learn, TensorFlow, PyTorch*

- Deployed a personal engineering portfolio featuring *RazaMind AI*, an embedded assistant answering visitor questions about projects; built a multi-tool ReAct agent and a reusable FastAPI/Tortoise ORM authentication service.
- Published 10+ end-to-end ML/DL projects: CIFAR-10 with ResNet50, face mask detection, cat-vs-dog and Fashion-MNIST CNNs, MNIST digits, breast cancer and heart disease prediction, fake news detection, sentiment analysis, credit card fraud detection, and Titanic survival modelling.

PROFESSIONAL EXPERIENCE

Machine Learning Engineer Intern | Developers Hub Corporation (Remote) *2025*

- Built an end-to-end IMDB sentiment system using TF-IDF features, linear classifiers, and a Streamlit interface; developed ML workflows for NLP, classification, and clustering with reproducible prediction pipelines.

Machine Learning Engineer Intern | Code Alpha (Remote) *2025*

- Delivered projects in heart-disease prediction, agricultural image recognition, and speech-emotion recognition, training logistic-regression, CNN, and LSTM models with structured preprocessing and performance analysis.

EDUCATION

BS Software Engineering | The Islamia University of Bahawalpur *2023-2027*

FSc Pre-Engineering | KIPS College *2021-2023*

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Automation & CRM:** Delivered a production-ready GoHighLevel (GHL) snapshot system covering CRM structure, funnels, workflows, Conversation AI, and compliance; completed n8n and Make.com training for agentic AI and business automation pipelines.
- Generative AI & LLM Engineering:** Completed CampusX coursework in LangChain, RAG pipelines, and ReAct agents.
- ML/DL & Backend Engineering:** Completed Hope to Skill and Siddhardhan ML/DL tracks, plus FastAPI, Tortoise ORM, JWT, PostgreSQL, Redis, Celery, and Docker training.
- One Million Prompts:** Dubai Future Foundation and Dubai Centre for Artificial Intelligence.